Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 15 May 2026

Afternoon

Paper reference **8FM0/25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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1.

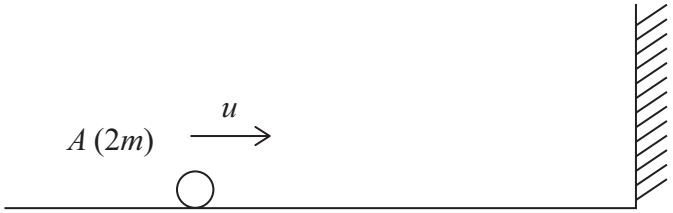


Figure 1

A particle A of mass $2m$ is at rest on a smooth horizontal floor. The particle A is projected along the floor with speed u towards a smooth vertical wall. The wall is fixed on the floor and is perpendicular to the direction of motion of A, as shown in Figure 1.

The particle A collides with the wall.

The coefficient of restitution between A and the wall is e.

(a) Find, in terms of m, u and e, the magnitude of the impulse received by A in its collision with the wall.

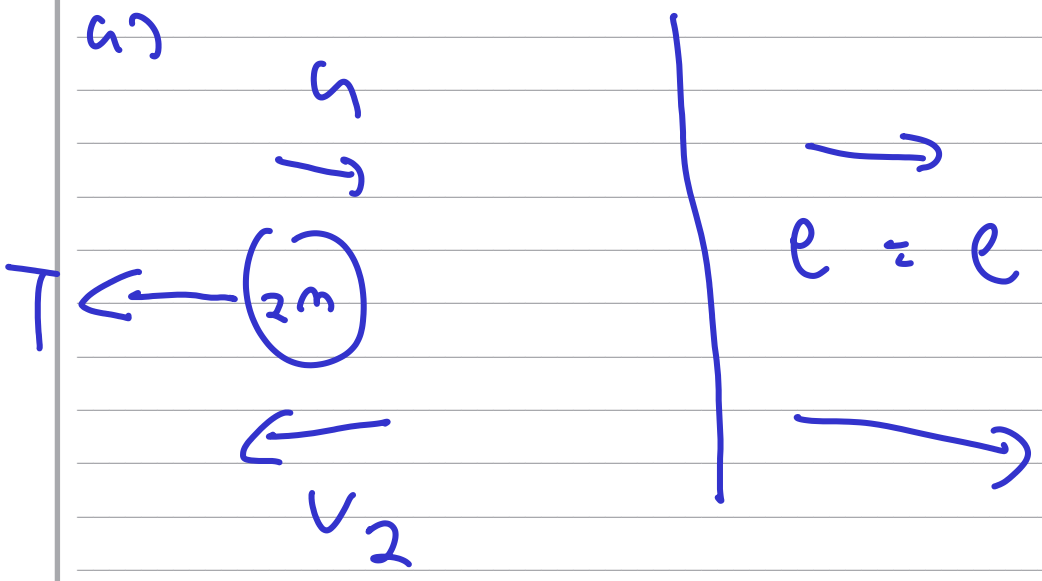
(3)

The kinetic energy lost by A in its collision with the wall is $\frac{16mu^2}{25}$

(b) Find the value of e.

(4)

a)



I = mv − mu

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Question 1 continued

using v

$$\frac{v_2}{4} = e$$

$$v_2 = 4e$$

$$k (\Leftarrow)$$

$$I = 2m(4e - (-4))$$

$$I = 2m(4e + 4)$$

$$I = 2m(4e + 1)$$

Change in ke

$$\frac{1}{2}m4^2 - \frac{1}{2}mv^2$$

$$\frac{1}{2} \times 2m4^2 - \frac{1}{2} \times 2m(4e)^2$$

$$\frac{16m^2}{25} = m4^2 - m4^2e^2$$

$$0.8 = \sqrt{\frac{16}{25}} = e$$

(Total for Question 1 is 7 marks)

2. A truck has mass 2100 kg. The maximum power of the engine of the truck is 24 kW.

Starting from rest, the maximum speed of the truck when it is moving along a straight horizontal road is S **km h**$^{-1}$

In a model of the motion of the truck

- the total resistance from non-gravitational forces **in all circumstances** has magnitude $\frac{192v}{5}$ newtons, where v m s^{-1} is the speed of the truck

Using the model

(a) find the value of S.

(5)

A straight road is inclined at an angle θ to the horizontal, where $\sin\theta = \frac{1}{49}$

The truck moves up the road with its engine working at maximum power.

At the instant when the acceleration of the truck **up the road** is $\frac{1}{175}$ m s^{-2}, the speed of the truck is V m s^{-1}

Using the model

(b) find the value of V.

(4)

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Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 9 marks)

3.

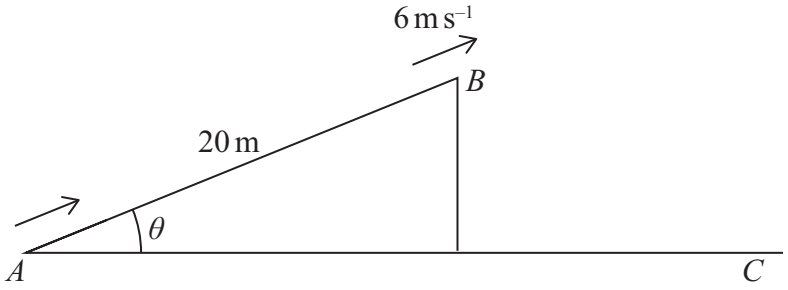


Figure 2

A rough ramp is fixed to horizontal ground.

The ramp is inclined to the ground at an angle θ, where $\sin\theta = \frac{1}{14}$

The point A is at the bottom of the ramp and the point B is at the top of the ramp.

The line AB is a line of greatest slope of the ramp and $AB = 20\,\text{m}$.

A small stone of mass 0.5 kg is projected from A directly towards B.
The stone reaches B with speed $6\,\text{m s}^{-1}$, as shown in Figure 2.

In an initial model

- the stone is modelled as a particle
- the total resistance to the motion of the stone from A to B, from non-gravitational forces, is modelled as a constant force of magnitude 0.45 N
- the initial speed of the stone is $U\,\text{m s}^{-1}$

Using the model and the work-energy principle,

(a) find the value of U.

(5)

In a second model

- the stone is modelled as a particle
- the total resistance to the motion of the stone from A to B, from non-gravitational forces, is modelled as being **zero**
- the initial speed of the stone is $W\,\text{m s}^{-1}$

(b) State whether W would be less than, equal to, or greater than U, giving a reason for your answer.

(2)

After leaving B, the stone hits the ground for the first time at the point C, shown in Figure 2, with speed $V\,\text{m s}^{-1}$

For the motion of the stone from B to C, the stone is modelled as a particle moving freely under gravity.

Using this model and **the principle of conservation of mechanical energy**

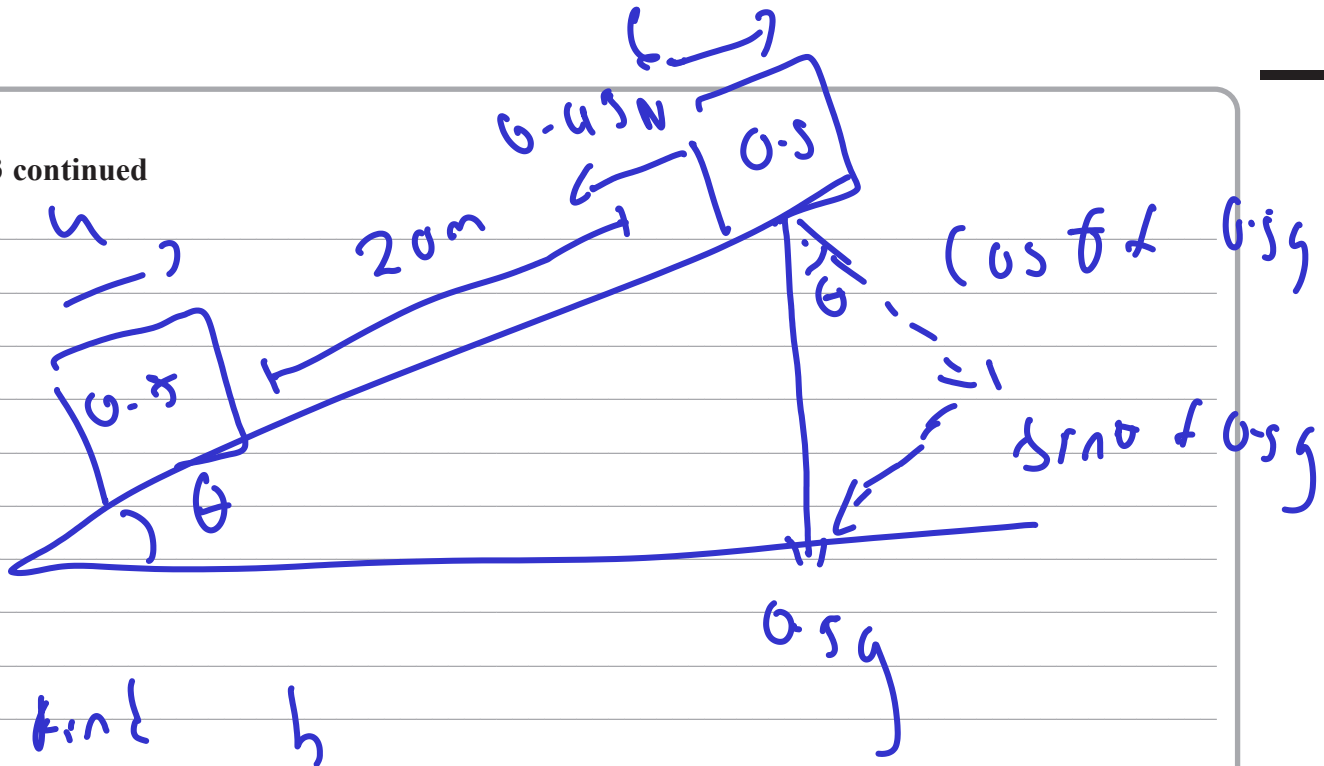
(c) find the value of V.

(4)

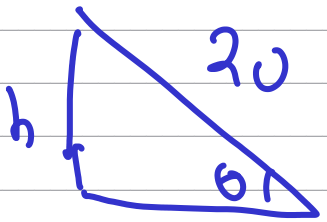
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Question 3 continued



to find h



$h = 20\sin\theta$

$h = 20 \times \frac{1}{5}$

$h = 4\text{m}$

$K = \frac{1}{2} \times 0.5 \times u^2$

$W = U$

$F = U$

$G = U$

$0.25u^2 = 196$

$u^2 = 784$

$u = 28\text{ms}^{-1}$

$K = \frac{1}{2} \times 0.5 \times v^2$

$W = 20 \times 0.49$

$E = 0$

$G = 0.5 \times g \times \frac{10}{7}$

$= 7$

Question 3 continued

b which is

Gas flow is less

resistance to electron

flow/ heat loss speed is quicker

C

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Question 3 continued

(Total for Question 3 is 11 marks)

4.



Figure 3

Two particles, P of mass m and Q of mass $2m$, are at rest on a smooth horizontal plane, as shown in Figure 3.

Particle P is projected with speed u along the plane towards Q and they collide.

The coefficient of restitution between P and Q is e, where $e < \frac{1}{2}$

(a) Show that the **speed** of P immediately after the collision is $\frac{u(1-2e)}{3}$, carefully justifying your reasoning.

(6)

(b) Find, in terms of u and e, the speed of Q immediately after the collision.

(1)

Particle Q goes on to hit a smooth vertical wall, also shown in Figure 3.

The wall is fixed on the plane and is perpendicular to the direction of motion of Q.

The coefficient of restitution between Q and the wall is $\frac{1}{4}$

The first collision between P and Q occurs at a distance d from the wall.

The second collision between P and Q occurs at a distance x from the wall.

Given that $e = \frac{1}{5}$

(c) find x in terms of d.

(6)

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Question 4 continued

Question 4 continued

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Question 4 continued

Question 4 continued

(Total for Question 4 is 13 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

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